

LEARNING RESOURCE GUIDE
ISC2 CCSP: Certified Cloud Security Professional (Aug 2026) | Professional Participant
Workbook

Data Location and Discovery Challenges in Cloud

1. Learning Objectives

- By the end of this lecture, you will be able to: Explain why cloud geographic distribution creates data location uncertainty and why this matters for regulatory compliance.
- By the end of this lecture, you will be able to: Distinguish between data residency and data sovereignty, and describe how these concepts interact in multi-region and multi-cloud environments.
- By the end of this lecture, you will be able to: Describe the specific discovery challenges created by cloud characteristics — automatic replication, tiering, data dispersion, and ephemeral data.
- By the end of this lecture, you will be able to: Explain how infrastructure tagging strategies and federated query models can address data location and compliance challenges.
- By the end of this lecture, you will be able to: Articulate the governance principle that data location must be explicitly configured and verified, not assumed.

2. Introduction

A European asset management firm with operations in Germany, France, and Singapore migrated its client portfolio data to a major cloud provider's European region. The IT team assumed — reasonably — that data stored in the EU region would remain within the EU. Eighteen months later, during a routine compliance audit, they discovered that the cloud provider's default cross-region replication for their database service had been quietly writing backup copies to a region in the United States. The configuration had been set during initial provisioning by an engineer who had not read the default settings documentation. The firm had been in violation of GDPR's data transfer restrictions for over a year without knowing it.

This is not a hypothetical. Variations of this scenario have occurred across regulated industries globally. Cloud providers design their default configurations for availability and performance — not for jurisdictional compliance. The customer is responsible for reading the documentation, applying the right constraints, and verifying that those constraints are actually working. In a multi-service, multi-account cloud environment, that is a non-trivial governance challenge.

Data location in cloud environments is not a static, observable fact. It is an outcome of deliberate configuration decisions. This lecture examines why location is uncertain in the cloud, what the legal consequences of that uncertainty are, and what technical and governance strategies exist to bring location under control.

3. Core Concepts Explained

3.1 Data Residency vs. Data Sovereignty

These two terms are frequently used interchangeably but describe distinct concepts that interact in important ways.

Data residency is the physical fact of where data is stored — the geographic location of the data centre or storage infrastructure that holds the data. An organisation can control data residency by selecting specific cloud regions and configuring services to write data only to those regions.

Data sovereignty is the legal concept of which country's laws govern the data based on where it physically resides. Data stored in Germany is governed by German law, including EU data protection regulation. Data stored in the United States may be subject to US law, including federal law enforcement access mechanisms under instruments such as the CLOUD Act, regardless of the nationality of the data owner or the data subjects. The same dataset, moved from one jurisdiction to another, can become subject to an entirely different legal regime — with different protection requirements, different breach notification obligations, and different government access provisions.

Business relevance: Organisations subject to GDPR, China's Personal Information Protection Law, India's Digital Personal Data Protection Act, or any national data localisation law must understand not just where they want their data to be, but where it actually is, what law governs it there, and whether transfers to other locations are permitted and under what conditions.

Practitioner angle: Data sovereignty claims become complicated when cloud infrastructure spans jurisdictions. A German company using a cloud provider's Frankfurt region might find that the provider's parent corporation is US-headquartered, raising questions about whether US government access requests could reach data stored in Frankfurt. This is an active area of legal and policy debate, and different cloud providers have adopted different contractual and technical positions in response.

3.2 Automatic Replication and Its Compliance Implications

Cloud providers replicate data by default for reliability. When an object is written to an S3 bucket, it may be stored redundantly across multiple availability zones within the same region. When a managed database service writes a transaction, it replicates to a standby instance in a different availability zone. When disaster recovery is configured, data is replicated to a secondary region. When edge caching is used, copies of data are distributed to edge locations around the world.

Each replication operation is a data transfer. Each transfer crosses geographic boundaries. Each destination jurisdiction may have different laws. The customer does not receive a notification for each replication event — it happens automatically, transparently, and continuously.

Major cloud providers do offer data residency controls — the ability to specify that data must stay within a defined geographic boundary: Google Cloud's Data Residency Commitments, Microsoft Azure's data residency controls, and AWS Region-locked services all provide

mechanisms to constrain replication. But these controls are opt-in, require deliberate configuration, and may not be available for all services within a cloud provider's portfolio. A cloud account with hundreds of services and workloads may have some services properly constrained and others operating under default cross-region configurations — and without continuous monitoring, the team may not know which is which.

3.3 Data Dispersion Within a Region

Even within a single cloud region, data does not necessarily reside in one physical location. Object storage services distribute large objects across multiple physical storage systems for performance and durability. A single large database table may be sharded across multiple physical nodes. A discovery scan that finds a data object in a particular region confirms that the logical record exists there, but the underlying physical distribution may span multiple data centre buildings within the region.

For most compliance purposes, the relevant unit is the logical region — not the physical building — and staying within a compliant region satisfies most jurisdictional requirements. However, in forensic and legal hold contexts, physical location matters: the ability to produce specific physical media as evidence, or to direct law enforcement to specific physical hardware, requires knowing not just the logical region but the physical infrastructure. Cloud providers generally do not disclose physical server locations to customers.

3.4 Ephemeral and Transient Data

Cloud environments generate transient data continuously: cache entries, log buffers, processing queues, lambda function memory, temporary files written by applications during data transformation, intermediate results in analytics pipelines. This data may contain sensitive information. It may persist longer than intended — an application failure before cleanup, a cache that was not properly configured to expire, a temporary file that was never deleted.

Transient data is almost never included in formal data inventories, which are built around persistent data stores. Standard discovery scanning may miss it entirely if it exists in memory, short-lived containers, or storage locations that are not within the scanning tool's scope. Governing transient data requires architectural discipline: designing systems to minimise sensitive data in transient states, using secure memory handling, and building cleanup logic that runs even on failure paths.

4. Comparison: Data Residency vs. Data Sovereignty

Attribute	Data Residency	Data Sovereignty
Definition	Physical location where data is stored	Which country's laws govern the data
Determined by	Cloud region selection and	Physical location of storage

	replication configuration	plus applicable treaties and laws
Controlled by	Customer configuration of cloud services	Primarily a legal concept; influenced by customer configuration
Changes when	Data is replicated or moved to a different region	Data crosses a national border (may change law that applies)
Key risk	Default cross-region replication violating residency requirements	Data in a jurisdiction with weaker protections or broader government access
Example regulation	GDPR Article 44 (transfers to third countries)	CLOUD Act (US government access to data held abroad by US companies)

5. Incident Analysis

Incident: The Default Replication Compliance Breach Pattern

Multiple GDPR enforcement actions in the 2020s have involved cloud-stored personal data being replicated outside the European Economic Area through default cloud provider settings that the data controller had not explicitly disabled. In a representative case examined by a European data protection authority, a mid-sized financial services organisation had documented an internal policy requiring that customer data remain within EU infrastructure. The policy was correctly described in their privacy documentation. What the policy did not address was the cloud provider's default disaster recovery replication, which wrote copies to a US region. The organisation's cloud team had provisioned the service by accepting the default configuration without reading the replication settings. No technical control verified that the policy requirement was met — the policy existed only as a document.

The enforcement outcome included a formal reprimand, a requirement to implement technical controls to enforce the residency policy, and a mandatory audit programme to verify compliance across all cloud services used by the organisation. The core finding was that a written policy is not a technical control — data residency requires configuration, monitoring, and ongoing verification, not documentation.

6. Memory Anchor

Think of it this way: Think of your data as water you pour into a container. In a traditional data centre, the container has clear walls and sits in a room you can see. In the cloud, the container is transparent but the pipes leading out of it are automatic, multiplying, and global — water flows to wherever the pipes are configured to send it, whether you intended it to or not. Controlling data location means explicitly sealing the outflow pipes to the locations you don't want, checking that the seals are holding, and monitoring the system continuously for new pipes you didn't know about.

7. Strategies for Managing Data Location

7.1 Cloud-Native Residency Controls

Most enterprise cloud providers now offer explicit data residency configuration. AWS provides region-specific service endpoints and supports Service Control Policies (SCPs) that restrict which regions services can be deployed in or which regions data can be written to. Azure offers data residency commitments for most services in its enterprise agreements. GCP provides Data Residency Commitments and Organisation Policy constraints that restrict resource creation to specified regions. These controls must be applied deliberately and their configuration must be documented and verified.

7.2 Infrastructure Tagging for Compliance Enforcement

Tagging cloud resources with their data classification level and applicable regulatory regime enables automated compliance verification. When every storage resource carries tags identifying the regulatory constraints that apply to the data it holds, automation can continuously check whether those resources are deployed in regions that satisfy those constraints. A resource tagged as holding GDPR-regulated data that is detected in a non-EU region triggers an alert — and potentially an automated remediation action. This converts data location compliance from a periodic audit activity into a real-time enforcement posture.

7.3 Federated Query Models

For analytical workloads that need to operate on regulated data, federated query approaches allow computation to run against data in its compliant location rather than moving the data to a central analytics environment. Rather than copying data from regulated storage to a central data lake — which would create a new, potentially non-compliant copy — queries execute in the regulated region and return only results. Cloud providers have built services around this pattern: AWS Athena federated queries, Google BigQuery Omni, and Azure Synapse Link all support querying data in its original location without requiring physical movement.

7.4 Multi-Cloud Discovery Challenges

Organisations operating across multiple cloud providers face a compounded discovery challenge. Each cloud provider has its own data residency controls, its own discovery tooling, and its own APIs. A discovery programme that works well within a single provider's ecosystem may not have visibility into data held by other providers. Cross-cloud discovery typically requires a third-party data governance platform capable of connecting to multiple providers simultaneously — or a federated approach where each provider's discovery tooling reports into a central governance system. Neither approach is trivial to implement at enterprise scale, and coverage gaps are common.

8. Exam Focus

Key areas tested by ISC2 CCSP in this topic:

- Know the distinction between data residency (physical location) and data sovereignty (legal jurisdiction that governs the data). Exam scenarios will test whether you can apply the right concept to a given situation.

- Understand that cloud providers' default configurations prioritise availability, not jurisdictional compliance — and that compliance requires explicit configuration by the customer.
- Know that automatic replication creates new copies in new jurisdictions, potentially without the customer's awareness, unless constrained.
- Understand the governance principle: data location is not self-enforcing. Policy must be backed by technical controls that are continuously monitored.
- Know that ephemeral and transient data may contain sensitive information and falls outside standard discovery approaches.
- Common distractor: exam questions may suggest that selecting a cloud region is sufficient to ensure data residency compliance. It is not — replication, caching, and backup configurations must also be reviewed.

9. Key Terms Glossary

Data residency: The physical geographic location where data is stored; determined by which cloud regions and data centres hold the data.

Data sovereignty: The legal principle that data is subject to the laws of the jurisdiction in which it physically resides.

Cross-region replication: A cloud provider feature that automatically copies data to a different geographic region, typically for disaster recovery or availability purposes.

Data localisation: A legal requirement that data about residents of a country must be stored and processed within that country's borders.

GDPR Article 44: The GDPR provision governing transfers of personal data to third countries or international organisations, requiring adequate protection or appropriate safeguards.

CLOUD Act: The US Clarifying Lawful Overseas Use of Data Act, which extends the ability of US law enforcement to compel US-based cloud providers to produce data stored abroad.

Service Control Policy (SCP): An AWS organisations feature that sets guardrails on the actions that IAM users and roles can perform, including restricting deployments to specific regions.

Federated query: An analytical query that runs against data in its original location without requiring the data to be physically moved or copied to a central repository.

Data dispersion: The distribution of data across multiple physical locations within a cloud provider's infrastructure, including within a single region.

Ephemeral data: Data that is created temporarily during computation or processing, which may contain sensitive information but exists outside normal persistent storage and governance controls.

10. Quick Revision Checklist

- ☐ I can distinguish data residency from data sovereignty and explain how they interact.
- ☐ I understand why cloud default configurations may cause data to leave a jurisdiction without the customer's explicit decision.
- ☐ I can name at least two mechanisms cloud providers offer to constrain data to specific regions.
- ☐ I understand that automatic replication for disaster recovery creates additional data copies in potentially non-compliant locations.
- ☐ I can explain why data dispersion within a region matters for forensic contexts even if it is less relevant for general compliance.
- ☐ I know what ephemeral data is, why it is challenging to discover, and what architectural discipline is required to manage it.
- ☐ I can describe the infrastructure tagging strategy for enforcing location-based compliance controls.
- ☐ I understand federated query models and why they allow analytics on regulated data without violating residency requirements.
- ☐ I know that a policy document stating a residency requirement is not a technical control — configuration and monitoring are required.
- ☐ I can apply the governance principle: data location must be explicitly configured and continuously verified, not assumed.

11. Practice Questions

Question 1 (Scenario-Based)

A German pharmaceutical company stores clinical trial data in an AWS EU (Frankfurt) region. Internal policy requires that all clinical data remain within the European Economic Area. During a compliance audit, the auditor discovers that AWS RDS automated backups are being written to the US East (N. Virginia) region due to the default cross-region backup configuration that was accepted during provisioning. The organisation's privacy policy document correctly states the EEA residency requirement. Which of the following BEST describes the compliance failure?

A. The policy was insufficiently specific about which cloud regions are considered compliant.

→ Incorrect. The policy clearly states EEA residency. The failure is not in policy specificity but in the absence of technical controls to enforce it.

B. The organisation treated a documented policy as a technical control, failing to implement and verify configuration to enforce the residency requirement.

→ Correct. A policy stating a residency requirement has no effect on where data actually goes. Technical controls — region restrictions, SCP policies, replication configuration — are required to enforce it. The failure is the gap between the written policy and the actual cloud configuration.

C. AWS was in breach of contract by routing backups to a non-EEA region without consent.

→ Incorrect. Cloud providers apply default configurations that customers agree to in their service terms. The customer accepted the default cross-region backup configuration at provisioning. The responsibility for configuring compliant settings rests with the customer.

D. The auditor applied the wrong regulatory framework — GDPR does not restrict backup data location.

→ Incorrect. GDPR's transfer restrictions in Article 44 apply to all personal data, including data in backup copies. Backup status does not exempt data from transfer restrictions.

Question 2 (Scenario-Based)

A cloud architect is designing data residency controls for a financial services firm operating under regulations that require customer financial data to remain within Canada. The firm uses AWS and processes data in the Canada (Central) region. The architect must ensure that no data leaves Canada, including during disaster recovery operations. Which combination of controls BEST meets this requirement?

A. Configure all services in the Canada (Central) region and document the residency requirement in the firm's cloud governance policy.

→ Incorrect. Documenting the requirement is insufficient. Default replication settings and disaster recovery configurations can still move data outside Canada without explicit technical restrictions.

B. Apply AWS Service Control Policies (SCPs) restricting data writes to the Canada (Central) region, disable cross-region replication for all services, configure disaster recovery using multi-AZ within Canada (Central), and implement continuous monitoring to alert on any out-of-region data transfers.

→ Correct. This combination applies technical enforcement at multiple layers: SCPs prevent services from writing to other regions, explicit replication configuration prevents automatic cross-region copies, multi-AZ provides resilience within the compliant region, and monitoring provides ongoing verification. This is defence-in-depth applied to residency compliance.

C. Purchase a Reserved Instance commitment for the Canada (Central) region to ensure data priority in that region.

→ Incorrect. Reserved Instances are a billing and capacity commitment, not a data governance control. They have no effect on where data is stored or replicated.

D. Use AWS Local Zones within Canada to ensure data is stored close to Canadian users.

→ Incorrect. AWS Local Zones extend the nearest AWS region to a metropolitan area for latency purposes. Data in a Local Zone is still part of the parent region's infrastructure and does not provide additional residency guarantees beyond what the parent region offers.

Question 3 (Scenario-Based)

A multinational retailer operates data warehousing workloads on Google Cloud in the EU. The data science team needs to run analytics on EU customer purchasing data alongside US inventory data. Moving the EU customer data to the US for analytics would violate GDPR transfer restrictions. Which architectural approach allows analytics to proceed without violating residency requirements?

A. Anonymise the EU customer data before transferring it to the US analytics environment.

→ Partially valid but not the architecturally preferred answer. Anonymisation can enable transfers, but truly anonymised data must be irreversibly de-identified — a high bar. The approach also creates operational complexity and risk of re-identification if the anonymisation is imperfect.

B. Use a federated query approach that runs analytics queries against EU customer data in its EU location, joining with US inventory data via a cross-region query, without physically copying the EU data to the US.

→ Correct. Federated query models — supported by tools like BigQuery Omni and AWS Athena federated queries — allow queries to run against data in its compliant location. The query result (aggregate analytics output) moves, not the underlying personal data. This preserves residency compliance while enabling cross-region analytics.

C. Transfer all EU data to the US under a Data Processing Agreement and conduct analytics from the US.

→ Incorrect. A DPA governs the relationship between controller and processor but does not by itself authorise transfers to the US. Under GDPR, transfers to the US require appropriate safeguards (Standard Contractual Clauses, adequacy decision, binding corporate rules). More importantly, this approach moves the data — which is exactly the outcome the question asks to avoid.

D. Run analytics only on US inventory data and exclude EU customer data from all analytical workloads.

→ Incorrect. Excluding the EU data from analytics defeats the purpose of the analytics programme. The requirement is to find an approach that allows analytics while preserving compliance — not to abandon the analytics.

Question 4

What is the primary difference between data residency and data sovereignty?

A. Data residency refers to cloud-stored data; data sovereignty refers to on-premises data.

→ Incorrect. Both concepts apply regardless of where data is stored — cloud, on-premises, or hybrid. The distinction is between physical location and legal governance, not storage type.

B. Data residency is the physical fact of where data is stored; data sovereignty is the legal concept of which jurisdiction's laws govern the data based on that location.

→ Correct. These are the precise definitions. Residency is observable physical fact; sovereignty is the legal consequence of that fact. The same data can be in residency-compliant storage (in the right country) but still subject to the laws of a second country if the cloud provider is domiciled there.

C. Data residency applies to personal data; data sovereignty applies to government data.

→ Incorrect. Both concepts apply to all data categories. The type of data (personal, commercial, governmental) does not determine which concept applies.

D. Data residency is controlled by the cloud provider; data sovereignty is controlled by the customer.

→ Incorrect. Data residency is controlled by customer configuration choices, not the cloud provider's defaults. Data sovereignty is a legal concept that follows physical location — neither party 'controls' it in the sense meant by this option.

Question 5

Which of the following cloud characteristics creates the GREATEST challenge for maintaining data residency compliance?

A. Cloud providers charge for data transfer between regions.

→ Incorrect. Transfer costs are an operational consideration, not a compliance challenge. Egress pricing does not prevent data from being transferred — it may even incentivise keeping data in one region for cost reasons, but this is incidental.

B. Cloud providers' default configurations include automatic replication across multiple regions for availability purposes, and these defaults are opt-in for compliance controls, not opt-out.

→ Correct. The default-availability-first design of cloud services means that without deliberate configuration, data replicates across regions. Customers must actively discover and configure compliance controls — the burden is on the customer to opt into compliant behaviour, not on the provider to make compliance the default.

C. Cloud providers do not disclose which physical data centres hold customer data.

→ Incorrect. This is a challenge for forensic contexts but not the primary residency compliance challenge. Most regulatory requirements are satisfied by logical region (country-level) residency, not physical data centre identification.

D. Cloud environments do not support encryption of data in transit between regions.

→ Incorrect. All major cloud providers encrypt data in transit by default, including inter-region traffic. This factual statement is incorrect, and even if it were true, encryption status does not determine residency compliance.

Question 6

Why is ephemeral data a particular challenge for data discovery and governance programmes?

A. Ephemeral data cannot contain sensitive information because it is processed in memory and never written to disk.

→ *Incorrect. Ephemeral data — cache entries, processing queues, lambda function memory, temporary files — can and does contain sensitive information. The fact that it is transient does not mean it is non-sensitive.*

B. Ephemeral data exists outside persistent storage and may be missed by standard discovery scanning tools, yet may contain sensitive information and persist longer than intended if cleanup logic fails.

→ *Correct. Standard discovery scans target persistent data stores. Ephemeral data in memory, short-lived containers, temporary storage, or processing queues is typically outside scanning scope. Applications can leave sensitive data in transient states if cleanup logic fails on error paths.*

C. Ephemeral data is automatically deleted by cloud providers after 24 hours, creating a narrow window for discovery.

→ *Incorrect. There is no universal 24-hour deletion rule for ephemeral data. The persistence of transient data depends on application design and cloud service configuration. Some transient data persists indefinitely if not explicitly cleaned up.*

D. Ephemeral data is encrypted by default and therefore exempt from data discovery requirements.

→ *Incorrect. Encryption status does not exempt data from discovery and governance requirements. Discovering that sensitive data exists in a location is separate from whether that data is encrypted.*

Question 7

An organisation's cloud governance team discovers that three cloud services are writing data to non-compliant regions despite a written policy prohibiting it. What is the MOST effective long-term control to prevent this from recurring?

A. Update the written policy to include specific cloud service names and prohibited regions.

→ *Incorrect. Updating documentation does not prevent cloud services from writing to non-compliant regions. The original policy already stated the requirement. The failure was in technical enforcement, not policy specificity.*

B. Implement infrastructure-as-code templates that encode compliant region configurations, combined with Service Control Policies that prevent resource creation in non-compliant regions, and continuous compliance monitoring that alerts on violations.

→ *Correct. This addresses the root cause — the gap between policy intent and technical configuration — with preventive controls (SCPs, approved templates) and detective controls (continuous monitoring). No individual service can be configured outside compliant regions without triggering an alert or being blocked outright.*

C. Require project teams to submit configuration change requests for review before any cloud service is provisioned.

→ Incorrect. Manual review processes are error-prone, create delays, and do not scale to the pace of cloud provisioning. They also fail on existing services that are modified after initial provisioning.

D. Conduct quarterly audits to identify non-compliant region configurations and remediate findings.

→ Incorrect. Quarterly audits are detective controls that only find violations after they have occurred and persisted for up to three months. They do not prevent violations and are insufficient as the primary control for a known recurring failure.

Question 8

What does the US CLOUD Act enable that is relevant to data sovereignty considerations for cloud customers?

A. It enables cloud providers to store US government data in non-US regions.

→ Incorrect. The CLOUD Act addresses law enforcement access to data, not US government data storage locations.

B. It enables US law enforcement authorities to compel US-headquartered cloud providers to produce data stored abroad, potentially affecting data sovereignty even when data is stored in compliant foreign regions.

→ Correct. The CLOUD Act extended existing US law to allow US law enforcement to compel production of data held by US-based providers regardless of where the data is physically stored. This means data stored in the EU on a US provider's infrastructure may be accessible to US government requests, which some European data protection authorities have cited as a data sovereignty concern.

C. It requires cloud providers to encrypt all customer data stored in the United States.

→ Incorrect. The CLOUD Act does not impose encryption requirements. It is a law enforcement access statute.

D. It prohibits cloud providers from replicating US citizen data outside US borders.

→ Incorrect. The CLOUD Act does not impose data localisation requirements. It is about law enforcement access, not data movement restrictions.

Question 9

Which discovery challenge is UNIQUE to multi-cloud environments compared to single-cloud deployments?

A. Sensitive data may exist in multiple data types — structured, unstructured, and semi-structured.

→ Incorrect. All three data types exist regardless of whether an organisation uses one cloud provider or many. This is not unique to multi-cloud.

B. No single discovery tool natively spans all cloud providers, requiring either multiple provider-specific tools or a third-party governance platform to maintain cross-environment visibility.

→ *Correct. Each cloud provider's native discovery tools are designed to operate within that provider's ecosystem. In a multi-cloud environment, a discovery programme must either deploy and integrate multiple provider-specific tools or adopt a third-party platform capable of connecting to all providers. Either approach introduces integration complexity and coverage gaps.*

C. Multi-cloud environments are subject to different encryption standards in each cloud.

→ *Incorrect. Encryption standards (AES-256, TLS, etc.) are consistent across major cloud providers. Differences in implementation do not create discovery challenges specific to the multi-cloud context.*

D. Data residency controls are not available from any cloud provider in multi-cloud configurations.

→ *Incorrect. Individual cloud providers offer their own residency controls regardless of whether the customer also uses other providers. Multi-cloud does not disable any provider's residency controls — it creates a governance coordination challenge, not a technical unavailability.*

Question 10

An organisation uses consistent tagging to mark all cloud storage resources with their data classification and applicable regulatory regime. A monitoring tool detects a storage bucket tagged as 'GDPR-regulated' in a non-EEA region. What type of security control does this monitoring represent?

A. Preventive control — it stops the bucket from being created in a non-compliant region.

→ *Incorrect. Monitoring that detects an existing misconfiguration is a detective control. A preventive control would block the bucket from being created in the non-compliant region in the first place (e.g., an SCP).*

B. Detective control — it identifies a residency violation after the fact so it can be remediated.

→ *Correct. The monitoring tool detects the violation after the resource has been created. Detection enables remediation but does not prevent the violation. To distinguish: an SCP that blocks creation is preventive; an alert that fires when a tagged resource appears in the wrong region is detective.*

C. Corrective control — it automatically moves the data to a compliant region.

→ *Incorrect. A corrective control takes remedial action. Monitoring that only alerts is detective. If the monitoring were paired with automatic remediation (deleting or migrating the bucket), the combined system would include a corrective component.*

D. Compensating control — it replaces a missing preventive control.

→ *Partially accurate in concept but not the best answer. While detective monitoring can serve as a compensating control when preventive controls are missing, the question asks what type of control the monitoring itself represents, which is detective.*

